

NET731 — Advanced Networking

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20240429
Practical Instruction Sheet |
Week 3

CTU Training Solutions

Module	NET731 — Advanced Networking
Practical Title	Azure Virtual Machine Deployment
Week / Period	Week 3 20–26 March 2025
Total Marks	10
Submission	GitHub repository (screenshots + documentation)

3.1 Virtual Machine Deployment

Microsoft Azure | Upgrade | Search resources, services, and docs (Ctrl+G) | Copilot | guggles_queen14@ictou... | JOURNAL1 DIRECT TONY (05664556...)

Home

Create a virtual machine

Help me choose the right VM size for my workload | Help me create a low cost VM | Help me create a VM optimized for high availability

Basics | Disks | Networking | Management | Monitoring | Advanced | Tags | Review + create

Create a virtual machine that runs Linux or Windows. Select an image from Azure marketplace or use your own customized image. Complete the Basics tab then Review + create to provision a virtual machine with default parameters or review each tab for full customization. [Learn more](#)

Project details
Select the subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all your resources.

Subscription *

Resource group * [Create new](#)

Instance details

Virtual machine name *

Region * [Deploy to an Azure Extended Zone](#)

Availability options

Security type

Image * [See all images](#) [Configure VM generation](#)

☒ This image is compatible with additional security features. [Click here to swap to the Trusted launch security type.](#)

VM architecture ☐ Arm64 ☒ x64

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DEFAULT DIRECTORY (GIGGLES1K)

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You are in the free trial period. Costs associated with this VM can be covered by any remaining credits on your subscription. [Learn more](#)

Size *
[See all sizes](#) [Customize cores](#)

Enable Hibernation ☐

Hibernate is not supported by the size that you have selected. Choose a size that is compatible with Hibernation to enable this feature. [Learn more](#)

Administrator account

Authentication type ☐ SSH public key ☒ Password

Username * ✓

Password * ✓

Confirm password * ✓

Inbound port rules

Select which virtual machine network ports are accessible from the public internet. You can specify more limited or granular network access on the Networking tab.

Public inbound ports ☐ None ☒ Allow selected ports

Select inbound ports *

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Azure VMs have one operating system disk and a temporary disk for short-term storage. You can attach additional data disks. The size of the VM determines the type of storage you can use and the number of data disks allowed. [Learn more](#)

There is a charge for the underlying storage resources consumed by your virtual machine. [Learn more](#)

VM disk encryption

Azure disk storage encryption automatically encrypts your data stored on Azure managed disks (OS and data disks) at rest by default when persisting it to the cloud.

Encryption at host ☐

Encryption at host is not registered for the selected subscription. [Learn more](#)

OS disk

OS disk size

OS disk type *

Delete with VM ☒

Key management

Enable Ultra Disk compatibility ☐

Ultra disk is supported in Availability Zone(s) 1,2,3 for the selected VM size Standard_B2ats_v2.

Data disks for TechBridge-Web01

You can add and configure additional data disks for your virtual machine or attach existing disks. This VM also comes with a temporary disk.

LUN	Name	Size (GiB)	Disk type	Host caching	Delete with VM
Create and attach a new disk Attach an existing disk					

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Price

1 X Standard B2ats v2
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Subscription credits apply ⓘ
8.0122 USD/hr
[Pricing for other VM sizes](#)

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Name

Aiden Horn

Preferred e-mail address

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Preferred phone number

Basics

Subscription

Azure subscription 1

Resource group

TechBridge-RG

Virtual machine name

TechBridge-Web01

Region

South Africa North

Availability options

No infrastructure redundancy required

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Resource group

TechBridge-RG

Virtual machine name

TechBridge-Web01

Region

South Africa North

Availability options

No infrastructure redundancy required

Zone options

Self-selected zone

Security type

Standard

Image

Ubuntu Server 24.04 LTS - Gen2

VM architecture

x64

Size

Standard B2ats v2 (2 vcpus, 1 GiB memory)

Enable Hibernation

No

Authentication type

Password

Username

TechBridge

Azure Spot

No

Disks

OS disk size

Image default

OS disk type

Premium SSD LRS

Use managed disks

Yes

Delete OS disk with VM

Enabled

Ephemeral OS disk

No

Networking

Virtual network

TechBridge-VNet

Subnet

Public-Subnet (10.0.1.0/24)

Public IP

(new) TechBridge-Web01-ip

NIC network security group

None

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Validation passed

Where public IP and NAT, which VM is deleted

Management

Microsoft Defender for Cloud

System assigned managed identity

Login with Microsoft Entra ID

Auto-shutdown

Backup

Enable periodic assessment

Enable hotpatch

Patch orchestration options

Monitoring

Alerts

Boot diagnostics

Enable OS guest diagnostics

Enable application health monitoring

Advanced

Extensions

VM applications

Cloud init

User data

Disk controller type

Proximity placement group

Capacity reservation group

Basic (free)

Off

Off

Off

Disabled

Off

Off

Image Default

Off

On

Off

Off

None

None

No

No

SCSI

None

None

None

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Create a virtual machine

Help me create a VM optimized for high availability +2

Basics Disks Networking Management Monitoring Advanced Tags Review + create

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This subscription may not be eligible to deploy VMs of certain sizes in certain regions.

Project details

Select the subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all your resources.

Subscription * Azure subscription 1

Resource group * TechBridge-RG
[Create new](#)

Instance details

Virtual machine name * TechBridge-DB01

Region * (Africa) South Africa North
[Deploy to an Azure Extended Zone](#)

Availability options No infrastructure redundancy required

Security type Standard

Image * Ubuntu Server 24.04 LTS - x64 Gen2 (free services eligible)
[See all images](#) | [Configure VM generation](#)

This image is compatible with additional security features. [Click here to swap to the Trusted launch security type.](#)

VM architecture Arm64
x64

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Help me create a low cost VM

Help me choose the right VM size for my workload

Validation passed

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Price

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Preferred phone number

Basics

Subscription

Azure subscription 1

Resource group

TechBridge-RG

Virtual machine name

TechBridge-DB01

Region

South Africa North

Availability options

No infrastructure redundancy required

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Help me create a VM optimized for high availability

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Help me choose the right VM size for my workload

Validation passed

Zone options

Self-selected zone

Security type

Standard

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Enable Hibernation

No

Authentication type

Password

Username

TechBridge

Azure Spot

No

Disks

OS disk size

Image default

OS disk type

Premium SSD LRS

Use managed disks

Yes

Delete OS disk with VM

Enabled

Ephemeral OS disk

No

Networking

Virtual network

TechBridge-VNet

Subnet

Internal-Subnet (10.0.2.0/24)

Public IP

(new) TechBridge-DB01-ip

NIC network security group

None

Accelerated networking

On

Place this virtual machine behind an existing load balancing solution?

No

Delete public IP and NIC when VM is deleted

Disabled

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Create a virtual machine

Help me create a VM optimized for high availability

Help me create a low cost VM

Help me choose the right VM size for my workload

Validation passed

Where the public IP and DNS were VM ID deleted

Where the public IP and DNS were VM ID deleted

Management

Microsoft Defender for Cloud

System assigned managed identity

Login with Microsoft Entra ID

Auto-shutdown

Backup

Enable periodic assessment

Enable hotpatch

Patch orchestration options

Basic (free)

Off

Off

Off

Disabled

Off

Off

Image Default

Monitoring

Alerts

Boot diagnostics

Enable OS guest diagnostics

Enable application health monitoring

Off

On

Off

Off

Advanced

Extensions

VM applications

Cloud init

User data

Disk controller type

Proximity placement group

Capacity reservation group

None

None

No

No

SCSI

None

None

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Health monitoring

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Now it's possible to execute multiple scripts at the same time, manage their progress and persist Command feature, now available through Azure CLI and PowerShell.

Run Command uses the VM agent to let you run a script inside this virtual machine. This can be h Select a command below to see details.

Name	Description
RunShellScript	Executes a Lin
FleetDiagnosticsLinux	Run FleetDiag
ifconfig	List network c
ipcommand	Use the ip uti

Run Command Script

RunShellScript

Script execution complete

Run

Output

Enable succeeded:
[stdout]
PING 10.0.2.4 (10.0.2.4) 56(84) bytes of data.
64 bytes from 10.0.2.4: icmp_seq=1 ttl=64 time=1.05 ms
64 bytes from 10.0.2.4: icmp_seq=2 ttl=64 time=1.54 ms
64 bytes from 10.0.2.4: icmp_seq=3 ttl=64 time=1.91 ms
64 bytes from 10.0.2.4: icmp_seq=4 ttl=64 time=1.03 ms

--- 10.0.2.4 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3005ms
rtt min/avg/max/mdev = 1.033/1.381/1.910/0.366 ms
[stderr]

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